Department of Mathematics
Motilal Nehru National Institute of Technology Allahabad
Mathematics II (MAN12101)
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Assignment 6 (Probability and Statistics)

1. A problem of mathematics is given to three students A, B, C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved?

2. Using Baye's theorem solve the following: Rohit and Raman are two students and their chances of solving a problem in statistics correctly are $\frac{1}{6}$ and $\frac{1}{8}$ respectively. If the probability of their making a common error is $\frac{1}{525}$ and they obtain the same answers, then find the probability that their answer is correct.

3. Two players A and B play tennis games. Their chances of winning a game are in the ratio 3 : 2 respectively. Find A's chance of winning at least two games out of four games played.

4. A quiz consists of 20 MCQs, each with four options, out of which only one is correct. A person answers all the questions randomly. What is the probability that exactly 5 questions are answered correctly?

5. The number of traffic accidents per week in a small city follows a Poisson distribution with mean equal to 3. What is the probability that exactly 2 accidents occur in 2 weeks?

6. In four tosses of a coin, let X be the number of heads. Find the probability distribution of X and hence find the mean and variance of X.

7. A special dice is prepared such that the probabilities of throwing 1, 2, 3, 4, 5 and 6 points are:
$$\frac{1-k}{6}, \quad \frac{1+2k}{6}, \quad \frac{1-k}{6}, \quad \frac{1+k}{6}, \quad \frac{1-2k}{6}, \quad \frac{1+k}{6}$$
respectively. If two such dice are thrown, find the probability of getting a sum equal to 9.

8. If the density function of a continuous random variable X is given by
$$f(x) = \begin{cases} ax, & 0 \le x \le 2 \\ (4-x)a, & 2 \le x \le 4 \\ 0, & \text{otherwise} \end{cases}$$
   (a) Find the value of a.
   (b) Find the cumulative distribution function (CDF) of X.
   (c) Find $P(X > 2.5)$.

9. The probability density function for continuous random variable X is given by
$$f(x) = \begin{cases} \frac{1}{2}\sin x, & 0 \le x \le \pi \\ 0, & \text{otherwise} \end{cases}$$
Find the mean and variance.

10. A discrete random variable X has the following probability distribution:

X	0	1	2	3	4
$P(X)$	c	$2c$	$2c$	c^2	$5c^2$

   (a) Find the value of c.
   (b) Evaluate $P(X < 3)$ and $P(0 < X < 4)$.
   (c) Find the mean and variance of X.

11. If the density function of a continuous random variable X is given by
$$f(x) = ae^{-2|x|}, \quad -\infty < x < \infty$$

(a) Find the value of a.
(b) Find the standard deviation of the distribution.

12. The diameter of an electric cable, denoted by X, is assumed to be a continuous random variable with probability density function $f(x) = 6x(1-x), \quad 0 \leq x \leq 1$. Then:

    (a) Show that $f(x)$ is a probability density function.
    (b) Determine a number k such that $P(X < k) = P(X > k)$.

13. Find the median and mode from the following data:

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
No. of Students	2	18	30	45	35	20	6	3

14. Find the mean and variance of the Poisson distribution.

15. Define normal distribution and derive its mean and variance.

16. If X follows a normal distribution with mean 2 and variance 4, find $P(X > 2)$.

17. If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2000 individuals, more than two will get a bad reaction.

18. A manufacturer who produces medicine bottles finds that 0.1% of the bottles are defective. The bottles are packed in boxes, each containing 500 bottles. A drug manufacturer buys 100 boxes from the producer. Using Poisson distribution, find how many boxes will contain:

    (a) No defective bottle
    (b) Almost three defective bottles
    (c) At least two defective bottles

19. In a book of 600 pages, there are 60 typographical errors. Assuming Poisson law for the number of errors per page, find the probability that a randomly chosen 4 pages will contain no errors.

20. In an examination, the candidates are awarded the following grades depending on the marks scored by them: Distinction: $> 80\%$, First class: $60\% \leq$ marks $< 80\%$, Second class: $45\% \leq$ marks $< 60\%$, Third class: $30\% \leq$ marks $< 45\%$, Fail: $< 30\%$. It was found that 8% of the students failed and 8% have secured distinction. Find the average marks obtained by a candidate and deduce the percentage of students placed in the second class. Assume normal distribution of marks.

21. A petrol pump is supplied with petrol once a day. If its daily volume of sales X (in thousands of liters) is distributed by

$$f(x) = 5(1-x)^4, \quad 0 \leq x \leq 1.$$

    Then what must be the capacity of its tank in order that the probability that its supply will be exhausted in a given day should be 0.01?

22. Find the mean, variance and standard deviation of the binomial distribution. Determine the binomial distribution whose mean is 9 and whose standard deviation is $\frac{3}{2}$.

23. If X is a Poisson variate such that

$$P(X = 2) = 3P(X = 4) + 45P(X = 6),$$

    find the mean and variance of X.

24. In a test of 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life of 2040 hours and standard deviation of 60 hours. Estimate the number of bulbs to burn for:

    (a) more than 2150 hours
    (b) more than 1920 hours but less than 2160 hours